

ELI — Adaptive Inference Governor

ELI v2.4.70

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Adaptive Inference Governor — Status and Remaining Plan

Updated for v2.4.70. Most of the gaps this plan identified in June are closed; the two that need a governor (throughput-aware think decision and per-call wall-time projection) are still open. Status per gap below; no governor code exists yet.

Scope: make ELI’s reasoning and inference *policy* (think or no-think, token budget, pass count, recovery) a pure function of the runtime environment and the task, so one redistributed binary behaves correctly on any machine with any model and no per-machine tuning.

Origin: a 2026-06-17 session in which EXPLAIN_COGNITION_RUNTIME burned a 273 s empty compact-synthesis pass, then two 539 s empty standard-synthesis passes, before a two-stage chain-of-thought fallback (470 s and 502 s): about 38 minutes for one question. The working final answer was 4,071 characters (about 1,018 tokens) in 502 s, roughly 2 tokens per second.

0. The one-sentence problem

ELI adapts its inference to **memory** (VRAM, context length) and **model size**, but not to **measured compute speed** (tokens per second). On a slow machine the same prompt that works on the development box can produce multi-minute turns.

1. What ELI already adapts to (do not rebuild)

Capability	Where	What it adapts to
Model-agnostic resolution (no baked model)	gguf_inference.get_model_path	settings, environment
Chat family and template	cognition/model_identity.py, model_output_tokens.py	GGUF metadata (template, architecture), not the file name
Thinking-channel detection	gguf_inference._is_thinking_model → model_identity.is_thinking_model	the embedded chat template, then architecture; no name list
Trained context length	core/startup_hardware_optimizer, ELI_MODEL_TRAINING	ELI_MODEL_TRAINING, then real GGUF context_length, else 8192; no filename table
Free-VRAM-aware sizing, one fit calculation	core/hardware_profile.py (smart_fit_config, unified_fit_config)	free VRAM, RAM budget, fit priority
KV-cache and compute-buffer reserve	hardware_profile.py (_kv_cache_mb, _compute_graph_reserve_mb)	context, layers, batch
Graceful GPU-layer fallback	boot optimiser; runtime_snapshot.json	load success
Per-mode token budget	adaptive_load_report cognition/reasoning_modes.py	n_ctx (20 % of the window for quick, 30 % otherwise), prompt pressure, complexity, mode
Size tier → budget scale	detect_tier, tier_scale	model size
Live decode-speed EMA	model_tier.record_speed, measured_tok_s (fed from the non-stream generation path in gguf_inference)	measured tokens per second
Speed-capped pass count	model_tier.speed_passes	measured tokens per second (thresholds ELI_SLOW_TPS=5, ELI_FAST_TPS=15)
No-think on utility calls	gguf_inference._no_think_prefill, force_no_think()	structured output, small budgets (< 1024), quick mode
Redistribution-aware settings	core/runtime_settings.py	machine moves

The speed EMA is consumed in exactly one place, speed_passes(). Nothing in the think decision, the token budget or mode selection reads it. speed_passes() reduces how many full generations a mode runs and never caps output length.

2. The gaps and where each stands

Gap	Was	Status
A thinking detection was a filename allowlist	_is_thinking_model matched qwen3, deepseek-r1, ...	Closed. Detection reads the chat template, then the architecture.
B the think decision and token budget ignored throughput	no tok/s term in _no_think_prefill or the budget	Open. The inputs are structured, max_tokens < 1024, the quick mode, the force_no_think() scope and ELI_MODEL_THINK.

Gap	Was	Status
C no wall-time projection	nothing computes “at the live EMA this call takes about N seconds”	Open.
D empty output was not self-correcting	an unterminated <think> returned empty, then a full-price retry	Partly closed. An empty non-stream response is retried once with <code>force_no_think</code> in the engine (<code>_get_chat_response</code>), and an empty stream is retried once with <code>no-think</code> in <code>gguf_inference</code> . There is still no reserved answer budget.
E context sizing undershot the trained context	a filename table returned 32768 for <code>qwen3</code>	Closed. GGUF metadata is authoritative.
F <code>force_no_think</code> was a point patch	one call site	Partly closed. It now wraps compact grounded synthesis, the chain-of-thought stage prose, the correction repair, and the empty-response and stream retries in <code>_get_chat_response</code> . The standard synthesis fallback (<code>_synthesize_answer</code>) and coding-agent fix generation for <code>SELF_IMPROVE</code> are not wrapped.
G redundant proactive churn	<code>persona_updater</code> rebuilt the same overlay every tick	Closed. <code>Persona</code> and <code>user-profile</code> overlays are debounced (120 s) with a content dirty-check.

Note on gap G: `persona_updater` is deterministic (database reads and regexes); the cost was redundant CPU and SQLite work, not inference-lock contention.

3. Design for what remains: the Adaptive Inference Governor

A single policy object that every model call routes its *decision* through. It consumes `reasoning_modes`, `model_tier` and `hardware_profile`; it does not replace them.

3.1 Inputs (all available at runtime)

- **Model capability:** `model_identity.is_thinking_model(template, architecture)`, cached per model.
- **Throughput:** `model_tier.measured_tok_s()`, with a cold-start default from `detect_tier()` until the first measurement.
- **Context:** `n_ctx` from the runtime snapshot and `n_ctx_train` from metadata.
- **Task class:** below.

3.2 Task classes

1. **STRUCTURED:** JSON, routing, plan graphs. Never thinks (already true via `structured=True`).
2. **GROUNDING_SYNTHESIS:** evidence already gathered; the call only phrases it (the `EXPLAIN_*` actions, RAG answer synthesis, fix-patch rendering). Thinking adds nothing; force no-think.
3. **OPEN_REASONING:** genuine multi-step reasoning (chain of thought, tree of thoughts, constitutional, self-consistency, open chat on a hard question). Thinking is allowed but bounded by throughput.

3.3 The decision (a pure function)

```
policy(task_class, model_caps, tok_s, n_ctx, prompt_tokens) -> {  
  think: bool,  
  think_budget: int,      # max tokens allowed inside <think>  
  answer_budget: int,     # reserved tokens after </think>  
  passes: int,            # via speed_passes()  
}
```

- A model that does not think: think=False always.
- STRUCTURED or GROUNDED_SYNTHESIS: think=False.
- OPEN_REASONING: at or above ELI_FAST_TPS think with the full budget; between slow and fast scale think_budget so the projected wall time is within the target; at or below ELI_SLOW_TPS turn thinking off, or hard-cap think_budget.
- Always reserve answer_budget, so a thinking call cannot exhaust its budget before it answers.

3.4 Integration points

- gguf_inference._no_think_prefill: consult the governor instead of only structured, < 1024 and the environment; keep force_no_think() as the primitive and ELI_MODEL_THINK as the hard user override.
- reasoning_modes: split the target tokens into think_budget and answer_budget.
- Engine synthesis sites: tag_compact_grounded_synthesis and _synthesize_answer as GROUNDED_SYNTHESIS and the reasoning-mode runners as OPEN_REASONING.
- Feed the speed EMA from the stream path as well, so quick-mode chat updates it.

3.5 Configuration

ELI_SLOW_TPS and ELI_FAST_TPS exist (5 and 15). A soft per-turn wall-time target (ELI_TURN_LATENCY_TARGET_S) is proposed and does not exist yet. ELI_MODEL_THINK is the hard override. Nothing machine-specific is persisted.

4. Cross-machine behaviour to aim for

Machine and model	tok/s	grounded actions (EXPLAIN_*)	hard chat
8 GB GPU with a 30B MoE	about 2	no-think, one pass	bounded think, within the target, never empty
24 GB GPU with a 7B	about 40	no-think	full think, full budget
CPU-only with a 7B	about 3	no-think	think off or hard-capped, with an answer reserve
any machine with a non-thinking model	n/a	normal answer	normal answer
any machine with an unfamiliar reasoning model	any	detected through the template	handled

5. Phases

1. **Governor core** for grounded synthesis: the policy function, no-think wired through _no_think_prefill, coding-agent fix rendering tagged.
2. **Throughput-bounded open reasoning** with an answer reserve: split budgets in reasoning_modes.

3. **EMA from streaming** so quick mode updates the measured speed.

Each phase is independently shippable.

6. Verification

- Unit: `policy(...)` truth table over task class $\times$ throughput tier $\times$ model-thinks.
- No-model engine tests in the style of `tests/test*_no_gguf*.py`.
- Behavioural: `EXPLAIN_MEMORY_RUNTIME` and `EXPLAIN_COGNITION_RUNTIME` synthesis is non-empty on the first pass and runs no-think.
- Speed simulation: inject a fake `measured_tok_s()` and assert the think flag and budget flip.
- Existing guards that must stay green: `tests/test_gguf_think_stop_collision.py`, `tests/test_reasoning_mode_collision.py`, `tests/test_cot_think_budget.py`.

7. Risks and non-goals

- **Over-suppressing thinking** on hard turns at mid speed: only `GROUNDED_SYNTHESIS` is unconditionally no-think; `OPEN_REASONING` keeps thinking with a budget above slow.
- **Non-goals:** changing model selection, the agent set, persona content or the grounding contract; removing `ELI_MODEL_THINK`.